資料視覺化
Data Visualization

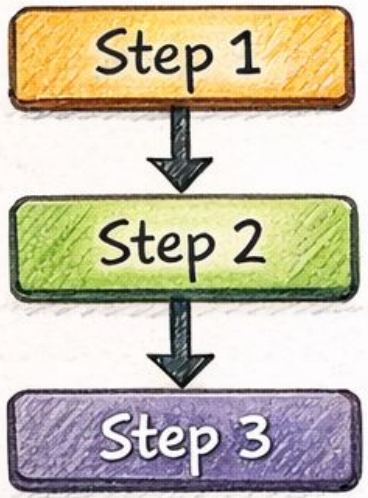
Week 2: Control Flows

The Structure of our Classes

- Lecture (me talking)
- Practical (me coding)
- Challenges (you coding)

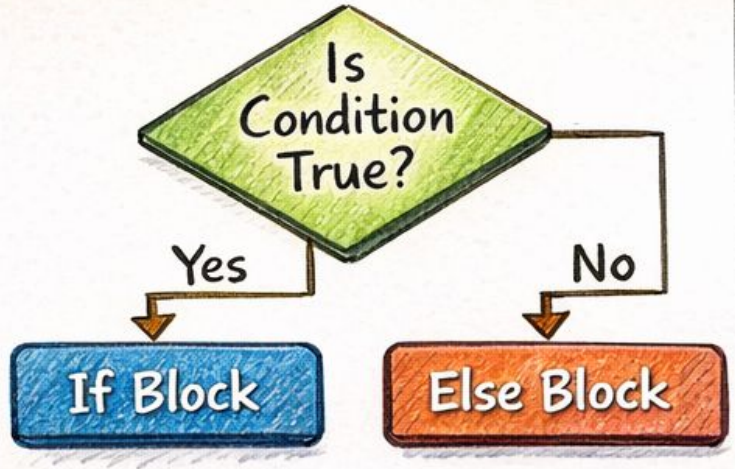
Python Control Loops

Sequential



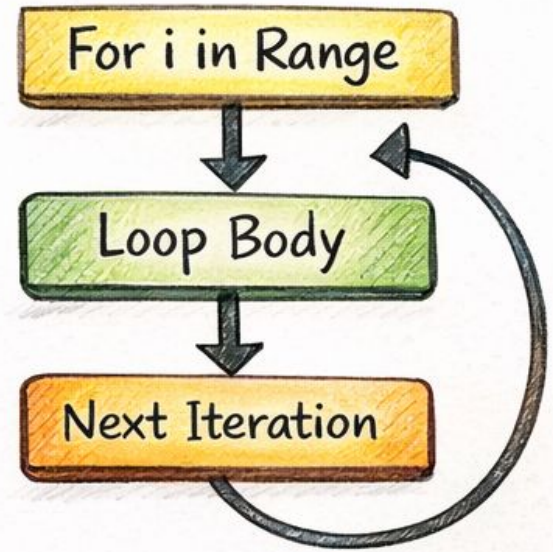
Execute in Order

Conditional (If – Else)



Choose Path Based on Condition

Iterative (For Loop)



Repeat Until Complete

Control Flows

- Sequential:
  Statements are executed one after the other in the order they appear.
  Note: we have been doing it the whole time
- Conditional:
  Statements are executed based on whether a condition is true or false (e.g., if, else, elif statements).
- Iterative:
  A set of statements is executed repeatedly as long as a condition is met or for a fixed number of times (e.g., for and while loops).

Condition Checks

- A condition is a statement that evaluates to either True or False.
- Examples:
  - x > 5
  - temperature < 30
  - name == "Alex”

Python Condition Operators

Comparison	Logical	Other
`==` equal	`and` both true	`is` identity
`!=` not equal	`or` one true	`is not` not same
`>` greater	`not` reverse	`in` membership
`<` less		`not in` not member
`>=` greater/equal		
`<=` less/equal		

Example:

```
if score >= 60 and name in students:
```

If... else...

```
if condition:
    print(“Condition is True”)
else:
    print(“Condition is not True”)
```

If... elif... else...

```
if condition 1:
    print(“Condition 1 is True”)
elif condition 2:
    print(“Condition 2 is True”)
else:
    print(“Condition 1 and 2 are not True”)
```

More about Python blocks

- A block is a group of statements that run together as a unit.
- Python uses indentation (exactly 4 spaces or 1 tab) to define blocks.
- Lines with the same indentation level belong to the same block.
- Blocks commonly appear after a colon :
- Used in structures like: if, for, while, etc.

For loop

```
names = ["Alice", "Bob", "Charlie"]
```

```
for name in names:
    print(name)
```

While loop

```
i = 0
While i < 6:
    print(i)
    i = i + 1
```

Note: While loop can keep on running forever

Time for Python